Fluid model forgetting case

Standalone theoretical note

This note presents the fluid model and some results.

Fluid bellman for fixed policy U

Recall U_ρ^1 corresponds to the value function when applying the fixed policy of using always product 1 (for simplicity of this note I'm going to use just U). For the purpose of this note we are going to work with the normalized coordinates $x = (1-\rho)S$. We start from the bellman equation

$$U(x) = (1 - D(x))\gamma U(x) + D(x)[R - c_1 + \gamma(p_1 U(T_\rho^s(x)) + (1-p_1)U(T_\rho^f(x)))] \tag{1}$$

Considering the transitions as

$$T_{dt}^s(x) = \rho^{dt}x + (1-\rho^{dt}) \tag{2}$$
$$T_{dt}^f(x) = \rho^{dt}x \tag{3}$$

We set

$$\rho^{dt} = e^{-\lambda dt} = 1 - \lambda dt + o(dt)$$
$$\gamma^{dt} = e^{-\delta dt} = 1 - \delta dt + o(dt)$$

Then, replacing in the transition functions we obtain

$$T_{dt}^s(x) - x = \lambda(1-x)dt + o(dt) \tag{4}$$
$$T_{dt}^f(x) - x = -\lambda x dt + o(dt) \tag{5}$$

These are our ODE equations

$$\begin{cases} \text{success:} & \dot{x} = \lambda(1-x) \\ \text{failure:} & \dot{x} = -\lambda x \end{cases} \tag{6}$$

Now, we note that the expected change of the state in the bellman equation would be

$$D(x)[p_1\lambda(1-x) + (1-p_1)(-\lambda x)]dt = D(x)\lambda(p_1 - x)dt \tag{7}$$

Hence we have

$$\dot{x} = D(x)\lambda(p_1 - x) \tag{8}$$

We can write our bellman equation over dt as follows

$$U(x) = (1 - D(x))\gamma^{dt}U(x) + D(x)[(R - c_1)dt + \gamma^{dt}(p_1 U(T_{dt}^s(x)) + (1-p_1)U(T_{dt}^f(x)))] \tag{9}$$

Now, we approximate $U(T_{dt}^s(x))$ and $U(T_{dt}^f(x))$ using Taylor expansion

$$U(T_{dt}^s(x)) = U(x) + \lambda(1-x)U'(x)dt + o(dt) \tag{10}$$
$$U(T_{dt}^f(x)) = U(x) - \lambda x U'(x)dt + o(dt) \tag{11}$$

replacing the discount factor's approximations

$$U(x) = (1 - D(x))(1 - \delta dt)U(x) + D(x)[(R - c_1)dt + \tag{12}$$

$$(1 - \delta dt)(U(x) + \lambda(p_1 - x)U'(x)dt)] + o(dt) \tag{13}$$

Rearranging terms we get

$$U(x) = U(x) + [D(x)(R - c_1) + D(x)(\lambda(p_1 - x)U'(x)) - \delta U(x)]dt + o(dt) \tag{14}$$

$$\Rightarrow 0 = [D(x)(R - c_1) + D(x)(\lambda(p_1 - x)U'(x)) - \delta U(x)]dt + o(dt) \tag{15}$$

Dividing by dt and taking $dt \to 0$ we get

$$\delta U(x) = D(x)(R - c_1) + D(x)\lambda(p_1 - x)U'(x) \tag{16}$$

To solve it, let $x(t)$ denote the fluid state at time t, starting from $x(0) = x$. The fluid Bellman equation can therefore be written as

$$\delta U(x(t)) = D(x(t))(R - c_1) + U'(x(t))\dot{x}(t). \tag{17}$$

By the chain rule,

$$\frac{d}{dt}U(x(t)) = U'(x(t))\dot{x}(t). \tag{18}$$

Hence,

$$\frac{d}{dt}U(x(t)) - \delta U(x(t)) = -D(x(t))(R - c_1) \tag{19}$$

Multiplying both sides by $e^{-\delta t}$,

$$e^{-\delta t}\frac{d}{dt}U(x(t)) - \delta e^{-\delta t}U(x(t)) = -(R - c_1)e^{-\delta t}D(x(t)). \tag{20}$$

The left-hand side is the derivative of $e^{-\delta t}U(x(t))$, so

$$\frac{d}{dt}\left[e^{-\delta t}U(x(t))\right] = -(R - c_1)e^{-\delta t}D(x(t)). \tag{21}$$

Integrating from 0 to T,

$$e^{-\delta T}U(x(T)) - U(x) = -(R - c_1)\int_0^T e^{-\delta t}D(x(t))\,dt. \tag{22}$$

Rearranging,

$$U(x) = e^{-\delta T}U(x(T)) + (R - c_1)\int_0^T e^{-\delta t}D(x(t))\,dt. \tag{23}$$

Since $D(x) > 0$, the sign of $\dot{x}(t)$ is determined by $p_1 - x(t)$. Thus, $x(t)$ moves toward p_1 from either side and

$$x(t) \to p_1 \qquad \text{as } t \to \infty. \tag{24}$$

Moreover, U is bounded and $\delta > 0$, so

$$e^{-\delta T}U(x(T)) \to 0. \tag{25}$$

Taking $T \to \infty$, we obtain the fluid value under the fixed policy of always using product 1:

$$U(x) = (R - c_1)\int_0^\infty e^{-\delta t}D(x(t))dt \tag{26}$$

where $x(t)$ solves

$$\dot{x}(t) = D(x(t))\lambda(p_1 - x(t)), \qquad x(0) = x. \tag{27}$$

This expression provides an integral representation of the fluid value function. The trajectory $x(t)$ is determined by the demand function $D(\cdot)$.

Fluid bellman for V

We start from the bellman equation

$$V(x) = (1 - D(x))\gamma V(x) + D(x) \max_{i \in \{1,2\}} \Big\{ R - c_i + \gamma [p_i V(T^s(x)) + (1 - p_i) V(T^f(x))] \Big\}$$

similar as before we can write

$$\begin{aligned} V(x) = & (1 - D(x))(1 - \delta dt) V(x) \\ & + D(x) \max_{i \in \{1,2\}} \big\{ (R - c_i) dt + (1 - \delta dt)[V(x) + \lambda (p_i - x) V'(x) dt] \big\} + o(dt) \end{aligned}$$

distributing and rearranging terms we get

$$\begin{aligned} V(x) = & (1 - D(x))(1 - \delta dt) V(x) + \\ & D(x) \max_{i \in \{1,2\}} \big\{ (R - c_i) dt + V(x) + [\lambda (p_i - x) V'(x) - \delta V(x)] dt \big\} + o(dt) \end{aligned}$$

Since $V(x)$ and $\delta V(x)$ do not depend on i we get

$$V(x) = V(x) + [D(x) \max_{i \in \{1,2\}} \big\{ R - c_i + \lambda (p_i - x) V'(x) \big\} - \delta V(x)] dt + o(dt)$$

dividing by dt and taking $dt \to 0$ we get

$$\delta V(x) = D(x) \max_{i \in \{1,2\}} \big\{ R - c_i + \lambda (p_i - x) V'(x) \big\}$$

We get that product 2 is optimal when

$$\begin{aligned} R - c_2 + \lambda (p_2 - x) V'(x) &> R - c_1 + \lambda (p_1 - x) V'(x) && (28) \\ \iff V'(x) &> \frac{\Delta c}{\lambda \Delta p} && (29) \end{aligned}$$

Which is basically the same equation we had in the original model.